

Methods: SDF1100.001

ALMA Thesis Planner (automated pipeline)

0.0.1 Phase 1: Data Archive Ingestion and Image Quality Assessment

1. Data Acquisition and Environment Setup:

- Ingest the pipeline-restored, science-ready FITS image products for ALMA Band 7 dataset ‘ADS/JAO.ALMA#2012.A.00028.S’ targeting the SDF1100.001 field from the ALMA Science Archive.
- Set up an analysis workspace using Python (with ‘astropy’, ‘spectral-cube’, and ‘photutils’) alongside CASA image utilities (‘imhead’, ‘imstat’, ‘imview’, ‘imfit’).

2. Data Product Verification:

- Inspect the primary-beam-corrected $1100\text{ }\mu\text{m}$ continuum FITS image and the 3D spectral line image cube covering the line redshift range $z = 5.63 - 5.72$.
- Confirm spatial and spectral parameters using ‘imhead’: spatial pixel scale, synthesized beam size ($\sim 1''$ spatial resolution, corresponding to ~ 6 kpc assuming $z = 5.65$), image footprint ($40'' \times 40''$, spanning ~ 240 kpc at an assumed redshift of $z = 5.65$), and channel width across the spectral window.
- Measure the $1\text{ }\sigma$ background noise level in emission-free regions of the $1100\text{ }\mu\text{m}$ continuum map (σ_{cont}) and in individual channel maps of the image cube (σ_{channel}) using CASA ‘imstat’.

0.0.2 Phase 2: Source Identification, Detection Verification, and 2D Spatial Mapping

1. Validation of SMA Candidates:

- Query the central sky coordinates of the 7 candidate [CII]-emitting galaxies previously reported from SMA observations in the $40'' \times 40''$ region.
- Extract localized $1100\text{ }\mu\text{m}$ continuum fluxes and spectral profiles at these 7 SMA positions to verify line and continuum detections at the higher ALMA sensitivity level.

2. Blind Source Searching for Faint Group Members:

- Conduct a systematic search for additional, fainter group members down to the ALMA sensitivity limit (approximately $10\times$ deeper than SMA in line emission and $7\times$ deeper in $1100\text{ }\mu\text{m}$ continuum emission).

- Search the 3D spectral cube for spatially and spectrally coherent emission line features exceeding a 3σ threshold across ≥ 2 adjacent channels using ‘spectral-cube’ and custom Python signal-extraction scripts.
- Apply an automated peak-finding routine (e.g., via ‘photutils’) on the $1100\text{ }\mu\text{m}$ continuum image to identify faint continuum sources down to a 3σ threshold.

3. Moment-0 Generation and Spatial Distribution Mapping:

- For every verified and newly identified source, collapse spectral channels spanning the line profile to generate integrated line intensity (Moment-0) maps using CASA ‘immoments’.
- Measure precise centroid positions (RA, Dec) for all detected sources by fitting 2D elliptical Gaussians to the Moment-0 and continuum images using CASA ‘imfit’.
- Plot the 2D projected spatial coordinates of all group members across the $40'' \times 40''$ field of view (~ 240 kpc at $z = 5.65$). Quantify spatial clustering and test for structural geometry (e.g., an elongated or filamentary spatial alignment) by computing spatial pair separations and positional angles.

0.0.3 Phase 3: 1D Spectral Extraction, Line Fitting, and Dynamic Profiling

1. 1D Spectrum Extraction:

- Define circular or beam-matched spatial extraction apertures centered on the 2D position of each identified source.
- Extract 1D spectra (flux density in mJy or Jy/beam versus observed frequency) across the spectral cube using Python ‘spectral-cube’.

2. Spectral Line Fitting:

- Fit single-component 1D Gaussian functions to the rest-frame [CII] line profiles using ‘scipy.optimize.curve_fit’ or CASA ‘specfit’.
- Derive key line parameters for each target:
- Observed central frequency (ν_{obs}) and corresponding spectroscopic redshift (z_{CII}).
- Integrated [CII] line flux ($S_{\text{CII}} \Delta \nu$ in Jy km s^{-1}).
- Line full-width at half-maximum (FWHM_{CII} in km s^{-1}).

3. Line-of-Sight Kinematics:

- Convert observed central frequencies into line-of-sight velocity offsets (Δv_{los} in km s^{-1}) relative to an assumed group systemic redshift of $z = 5.65$ using the relativistic Doppler relation:

$$\Delta v_{\text{los}} = c \cdot \frac{z_{\text{CII}} - 5.65}{1 + 5.65}$$

- Construct a velocity distribution histogram and evaluate the dynamic velocity spread across the group members to characterize line-of-sight velocity dispersion within the structure.

0.0.4 Phase 4: Continuum Photometry and Energetics ($L_{\text{CII}}/L_{\text{FIR}}$)

1. $1100\text{ }\mu\text{m}$ Continuum Photometry:

- Measure the $1100\text{ }\mu\text{m}$ continuum flux density ($S_{1100\text{ }\mu\text{m}}$) for each detected source using aperture photometry or 2D Gaussian fitting via CASA ‘imfit’.
- For targets without significant continuum emission ($<3\sigma$), calculate a 3σ upper limit as $3\sigma_{\text{cont}} \times \sqrt{\Omega_{\text{source}}}$.

2. [CII] Line Luminosity Calculation:

- Calculate rest-frame [CII] line luminosities (L_{CII} in $L_{\odot}$) from integrated line fluxes using the standard relation:

$L_{\text{CII}} = 1.04 \times 10^{-3} \times S_{\text{CII}} \Delta \nu \times \nu_{\text{rest}} \times D_L^2 \times (1 + z_{\text{CII}})^{-1}$ where $\nu_{\text{rest}} = 1900.536\text{ GHz}$ is the rest-frame [CII] frequency, and D_L is the luminosity distance in Mpc computed for z_{CII} assuming standard Flat Λ CDM cosmology ($H_0 = 70\text{ km s}^{-1}\text{ Mpc}^{-1}$, $\Omega_M = 0.3$).

3. Far-Infrared (FIR) Luminosity Estimation:

- Estimate total Far-Infrared luminosity (L_{FIR}), integrated over rest-frame $42.5 - 500\text{ }\mu\text{m}$ from single-band $1100\text{ }\mu\text{m}$ continuum fluxes.
- Adopt a standard modified blackbody (greybody) spectral energy distribution model:

$S_{\nu} \propto \nu^{\beta} B_{\nu}(T_{\text{dust}})$ under typical assumed dust parameters: dust temperature $T_{\text{dust}} \approx 40 - 50\text{ K}$ and dust emissivity index $\beta \approx 1.5 - 2.0$.

- Integrate the modeled greybody curve over rest-frame $42.5 - 500\text{ }\mu\text{m}$ to obtain L_{FIR} for each source (or 3σ upper limits where continuum is undetected). Propagate parameter ranges ($T_{\text{dust}} = 40\text{--}50\text{ K}$, $\beta = 1.5\text{--}2.0$) into systematic uncertainties on L_{FIR} .

4. $L_{\text{CII}}/L_{\text{FIR}}$ Energetics Analysis:

- Calculate the logarithmic luminosity ratio $\log_{10}(L_{\text{CII}}/L_{\text{FIR}})$ for every group member.
- Compare the derived individual and group-averaged ratios against the ratio range of $10^{-2.4}$ to $10^{-2.1}$ reported from prior SMA observations to evaluate dust heating properties, radiation field intensity, and star-formation energetics in this dense environment at $z \approx 5.65$.